

Foundation (Jalapeno)

- Q1.** Given $f(x) = x^2$ and $g(x) = 2x$, what is $f(g(x))$?
(A) $4x^2$
(B) $2x^2$
(C) x^2
(D) $4x$
(E) $2x$
- Q2.** If $f(x) = \sqrt{x}$ and $g(x) = x - 3$, what is the domain of $f(g(x))$?
(A) $[3, \infty)$
(B) $[-\infty, 3]$
(C) $[0, \infty)$
(D) $(-\infty, 3]$
(E) $[3, 0]$
- Q3.** For $f(x) = x + 2$ and $g(x) = 3x$, what is $f(g(2))$?
(A) 8
(B) 6
(C) 4
(D) 2
(E) 0
- Q4.** Given $f(x) = \frac{1}{x}$ and $g(x) = x + 1$, what is $f(g(x))$?
(A) $\frac{1}{x+1}$
(B) $\frac{x+1}{1}$
(C) $x + 1$
(D) $\frac{1}{x-1}$
(E) $x - 1$
- Q5.** If $f(x) = x^3$ and $g(x) = 2x - 1$, what is $f(g(2))$?
(A) 27
(B) 8
(C) 1
(D) 0
(E) -1
- Q6.** Given $f(x) = |x|$ and $g(x) = x - 2$, what is the range of $f(g(x))$?
(A) $[0, \infty)$
(B) $(-\infty, 0]$
(C) $[2, \infty)$
(D) $(-\infty, 2]$
(E) $[0, 2]$
- Q7.** For $f(x) = x^2 - 1$ and $g(x) = x + 3$, what is $f(g(-3))$?
(A) 8
(B) 0
(C) -1
(D) -8
(E) -9
- Q8.** Given $f(x) = \frac{x}{x+1}$ and $g(x) = 2x$, what is $f(g(x))$?
(A) $\frac{2x}{2x+1}$
(B) $\frac{2x}{2x-1}$
(C) $\frac{x}{x+1}$
(D) $\frac{x+1}{x}$
(E) $\frac{2x-1}{2x}$

Open Ended Questions (FRQ)

- Q9.** A bakery is having a sale on bread. The original price of a loaf is 2. The sale price can be represented by the function $f(x) = 2 - 0.5x$ where x is the number of days since the sale started. The cost of producing the bread can be represented by the function $g(x) = 0.5x + 1$. What is the profit function $f(g(x))$ and what is the profit after 4 days?

- Q10.** Given $f(x) = x^2$ and $g(x) = \sqrt{x}$, determine the domain and range of $f(g(x))$ and provide a real-world application of this composite function.

Chef's Tips

- Ensure students understand the concept of composite functions.
- Emphasize the importance of domain restrictions in composite functions.
- Provide real-world examples to help students interpret compositions.